Report 4.9 – Combinazioni di k con $TIMES = 10$

2 maggio 2026

1 Grafici dei risultati per k e $TIMES = 10$

I grafici mostrano rispettivamente l'andamento di:

- Avg fQ
- fQ Min Found
- Profit Found
- Weight Found
- Profit GAP
- Weight GAP
- n of fQ Min Found

al variare di $k \in \{1000, 2000, 3000, 4000, 5000\}$, confrontando i risultati usando QALS e NON utilizzando QALS.

I risultati riportati sono stati ottenuti utilizzando:

- $\lambda = \frac{\sum_{i=1}^{n} p_i}{3}$
- l'istanza `ksp_2.txt`
- `TIMES = 10`

Profit ottimo	7740
Weight ottimo	978
TIMES	10

Tabella 1: Soluzione di riferimento del problema di knapsack.

2 Tabella riassuntiva

k	QALS							NO QALS						
	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min
1000	749855518564.07	510388963914.0	11686	8152	-3946	-7174	1	-10200710258.33	-10200710854.33	6674	1001	1066	-23	2
2000	769098195352.37	481393338885.0	11315	7950	-3575	-6972	1	-10200710200.13	-10200710854.33	6674	1001	1066	-23	1
3000	832163777425.23	480968971551.0	11454	7947	-3714	-6969	1	-10200710619.53	-10200710854.33	6674	1001	1066	-23	6
4000	651092992544.7	457634883700.0	11780	7780	-4040	-6802	1	-10200710481.23	-10200710854.33	6674	1001	1066	-23	3
5000	740307028745.53	382521125146.0	13294	7212	-5554	-6234	1	-10200710420.33	-10200710854.33	6674	1001	1066	-23	2

Tabella 2: Risultati numerici per $k \in \{1000, 2000, 3000, 4000, 5000\}$ con `TIMES = 10`.

3 Grafici

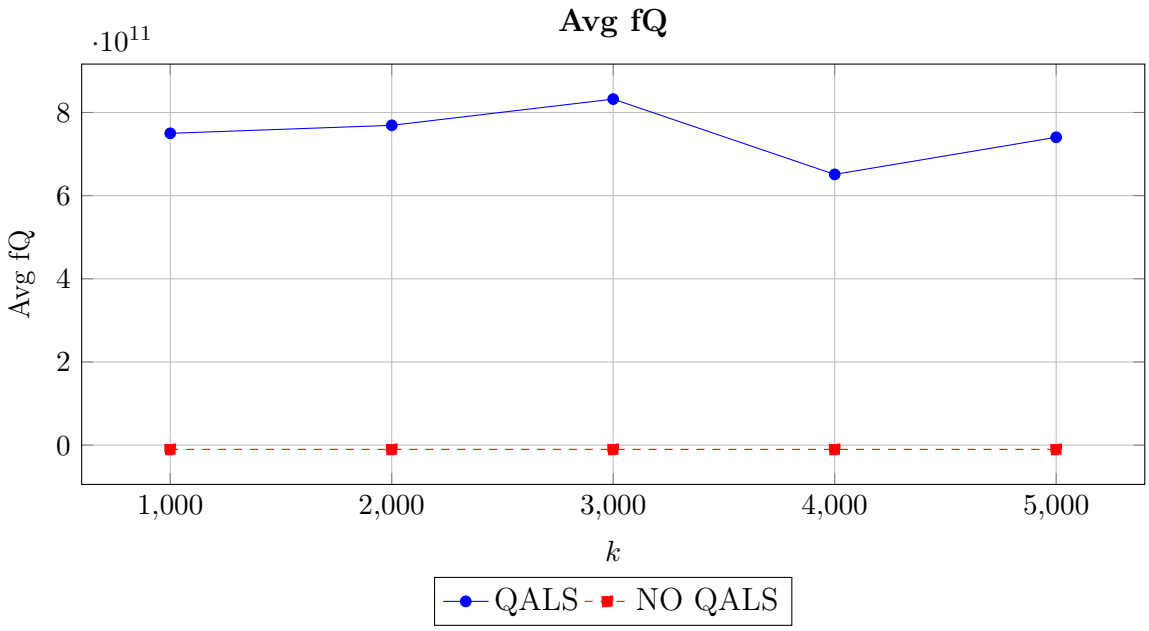

Figura 1: Andamento di Avg fQ al variare di k.

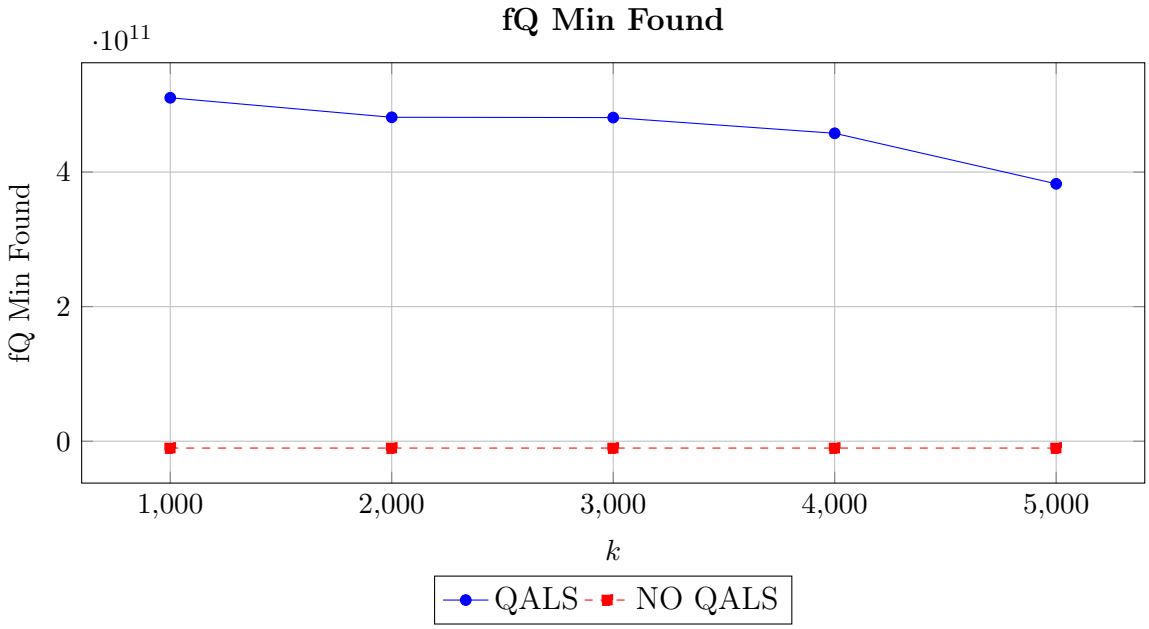

Figura 2: Andamento di fQ Min Found al variare di k.

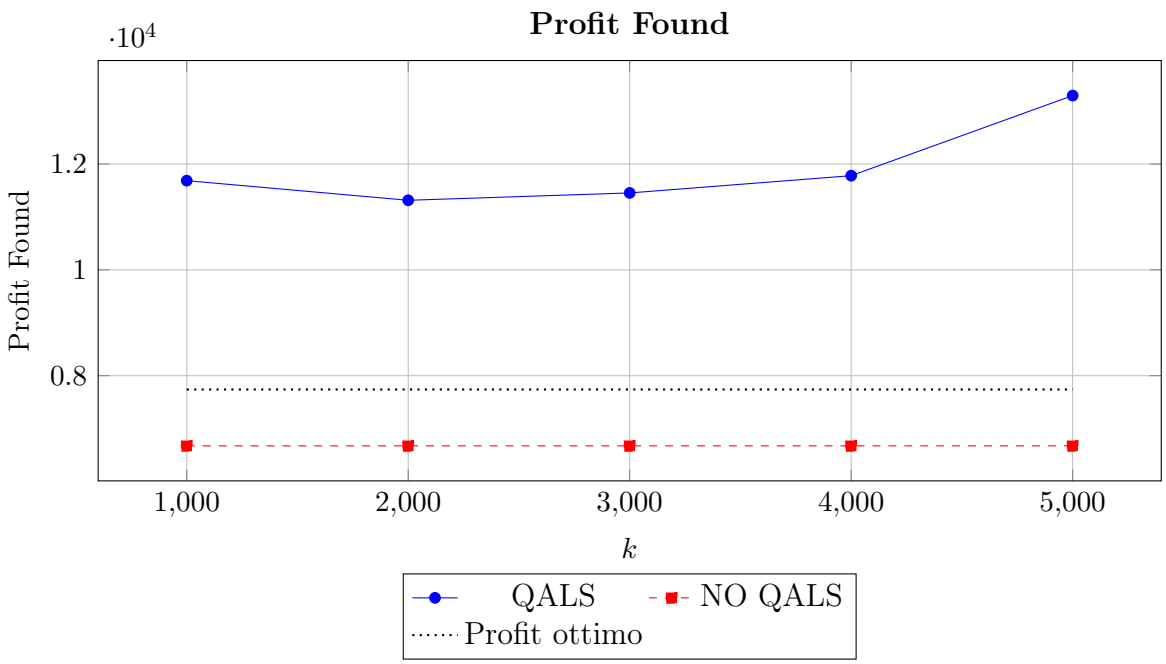

Figura 3: Andamento del profitto trovato al variare di k.

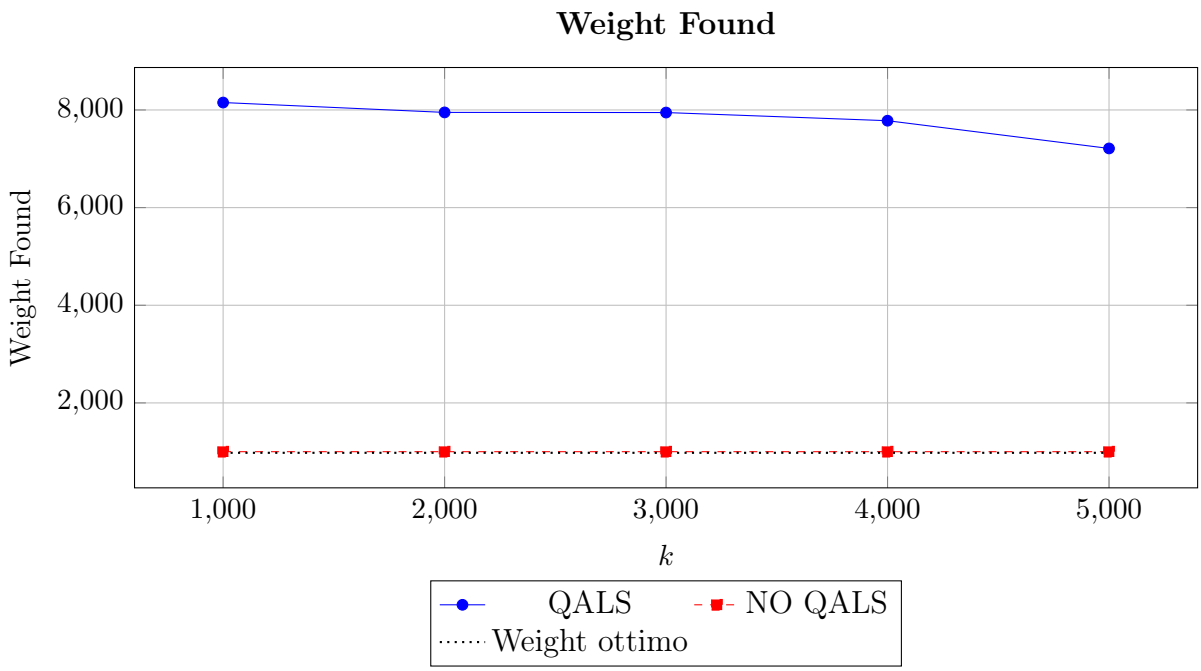

Figura 4: Andamento del peso trovato al variare di k.

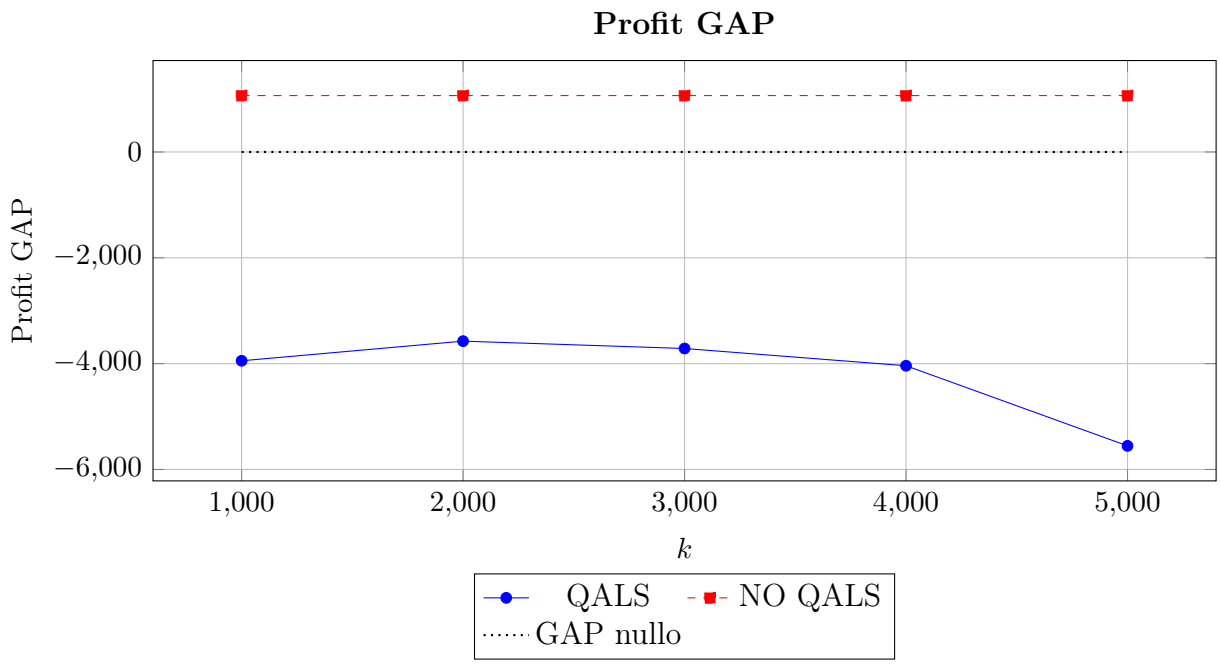

Figura 5: Andamento del Profit GAP al variare di k.

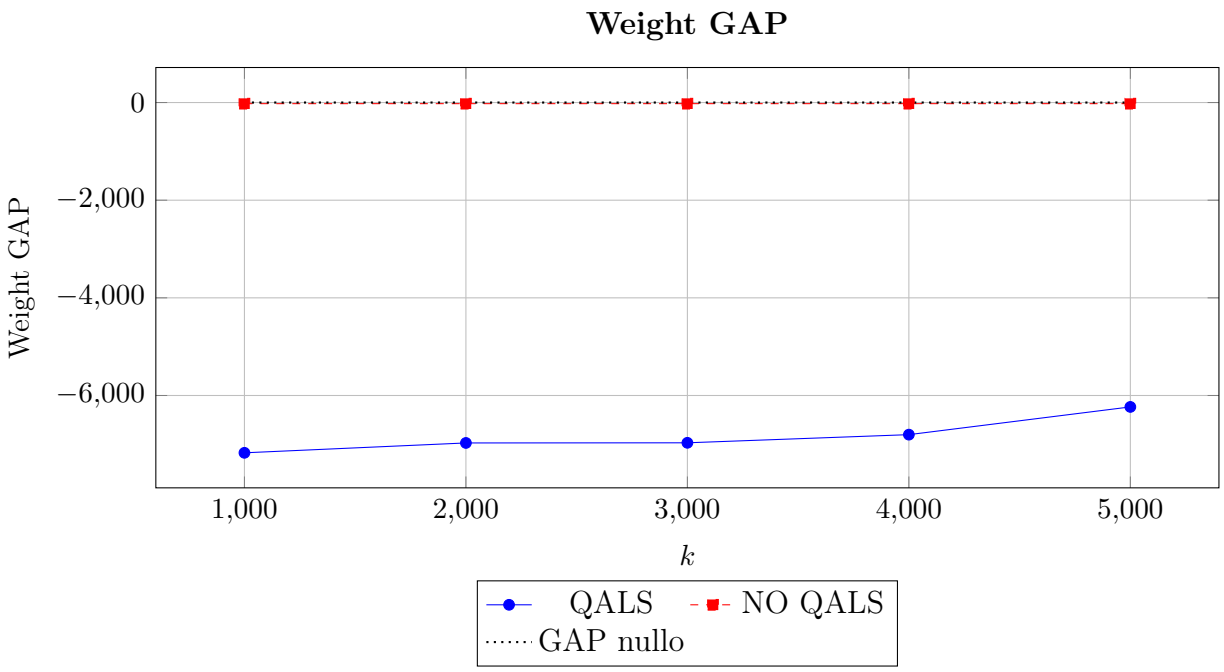

Figura 6: Andamento del Weight GAP al variare di k.

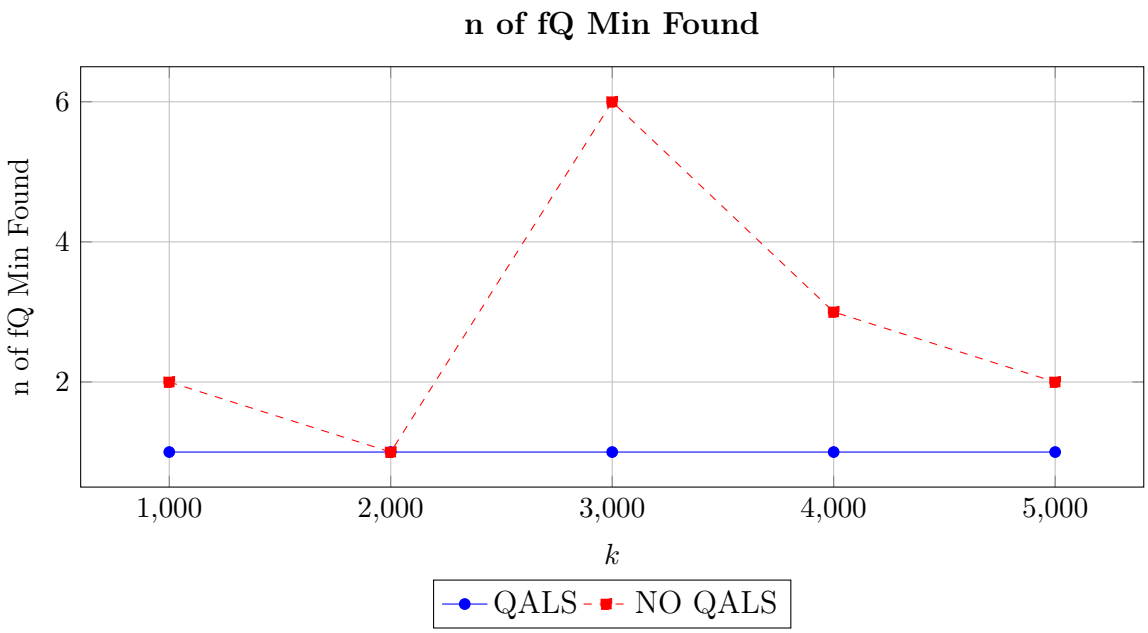

Figura 7: Numero di volte in cui è stato trovato il minimo fQ al variare di k.